

4. One type of car which is very efficient is a hybrid electric vehicle, which has both a conventional fuel engine and an electric motor which runs from batteries.

Data about a hybrid electric / petrol car is given below.

|                                                  |             |
|--------------------------------------------------|-------------|
| Minimum time taken to accelerate from 0 - 30 m/s | 12 s        |
| Mass of car                                      | 1 100 kg    |
| Mean CO <sub>2</sub> emissions                   | 90 g/km     |
| Mean fuel economy                                | 32 km/litre |

(a) The car travels 160 km per week.

(i) Calculate the mean mass of CO<sub>2</sub> emitted by the car in a week. [1]

CO<sub>2</sub> = ..... g

(ii) Calculate how many litres of fuel are used every week. [1]

Fuel used = ..... litres

(b) (i) Use data from the table and an equation from page 2 to calculate the maximum acceleration of the car. [3]

Acceleration = ..... m/s<sup>2</sup>



(ii) Use an equation from page 2 to calculate the size of the resultant force required to produce this acceleration. [2]

Resultant force = ..... N

(c) State **one** other way cars are made more energy efficient. [1]

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